NAKLAWI: An Inteligent Diagnosing and Knowlede Mobile Platform for Palm Care

GRADUATION PROJECT SHOWCASE 11
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1 Introduction

The agriculture and heritage of Saudi Arabia are largely dependent on date palm farming. which, has not been as lucrative lately because of the inability of majority of small-scale palm owners to take proper care of the trees. In order to solve this, we have come with Nekhlawi app that serves to diagnose diseases, prescribe treatment, and connect with specialists with the help of the AI. A survey was done to palm owners as well as agricultural experts to ascertain that the app has been designed to meet the needs of the user.

2 Problem Statement

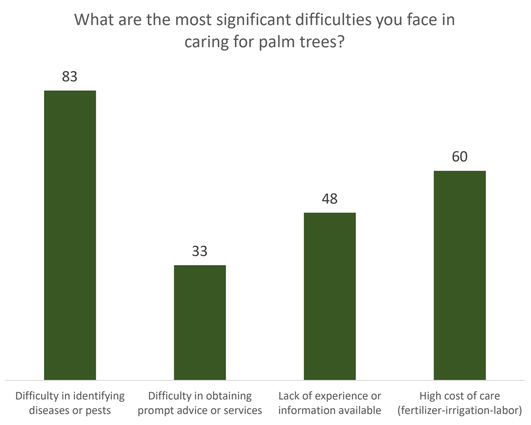
The owners of date palms are experiencing enormous challenges that put their farming sustainability under serious threat mainly because of several diseases that affect the production and the survival of palms resulting in an economic loss estimated to be 28 percent of the entire production. Moreover, the knowledge that is possessed by highly qualified farmers can easily be lost since there is no effective documentation. These unresolved problems show that there is an urgent demand to find innovative solutions to overcome the gap between the accessibility of technologies and the preservation of agriculture.

3 Gap Analysis

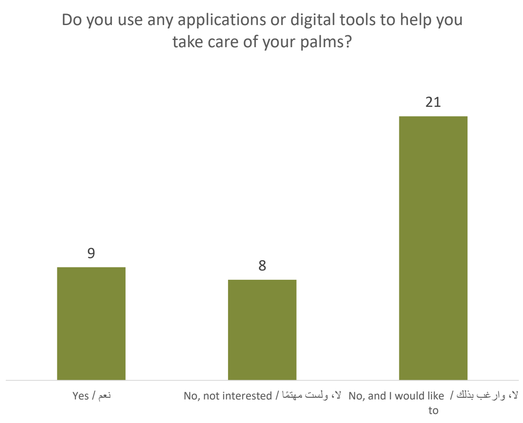
	AI Model	Mobile Support	Expert Consultations	Knowledge Hub	Diseases Diagnosis
NEKHLAWI	✓	✓	✓	✓	✓
DPXception [15]	✓	✓	✗	✗	✓
Web-Based Expert System [16]	✗	✗	✓	✓	✓
Palm tree disease detection using residual network [5]	✓	✗	✗	✗	✓
IoT-based system [17]	✓	✗	✗	✗	✓

4 Survey Results

To ensure the Nekhlawi App effectively addresses user needs, a comprehensive survey was conducted. The survey primarily focused on the following key questions:

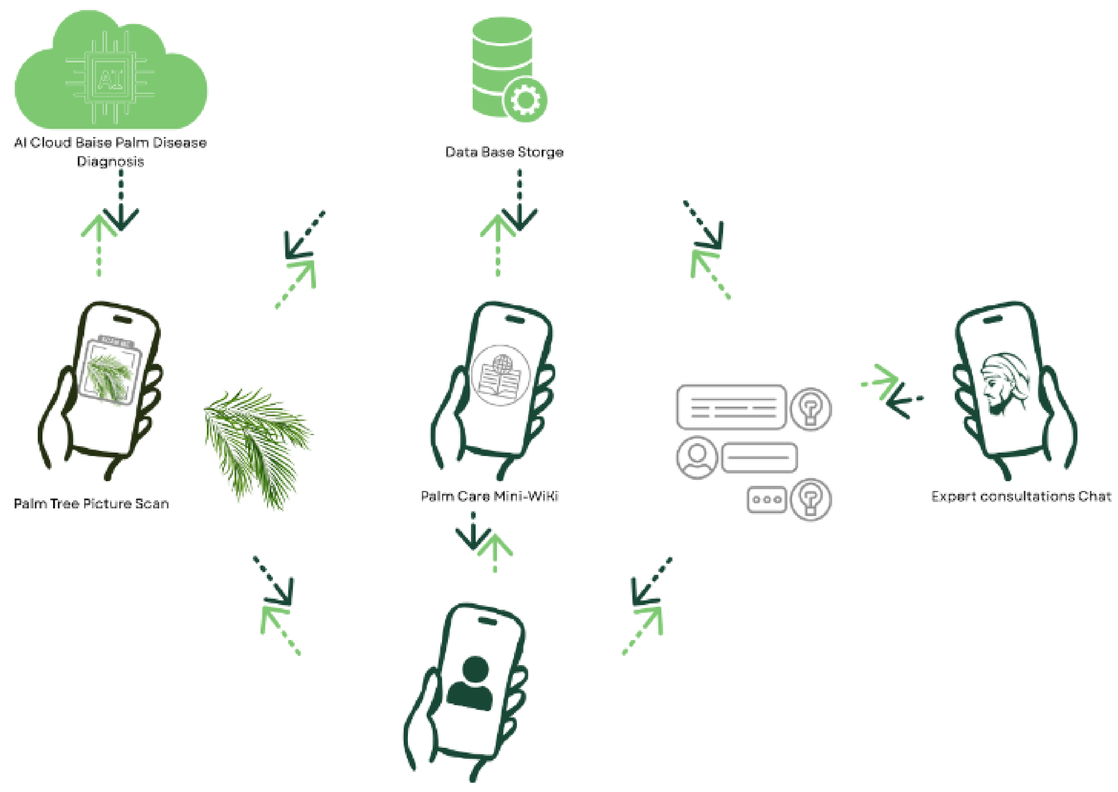


Users face challenges in palm tree disease diagnosis



High interest in using a palm tree care mobile application

5 System Overview



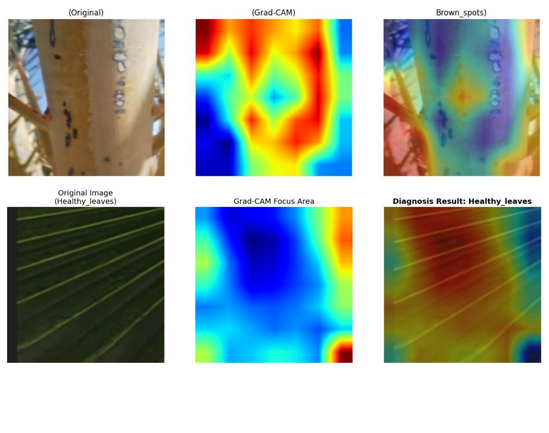
6 Proposed Solution

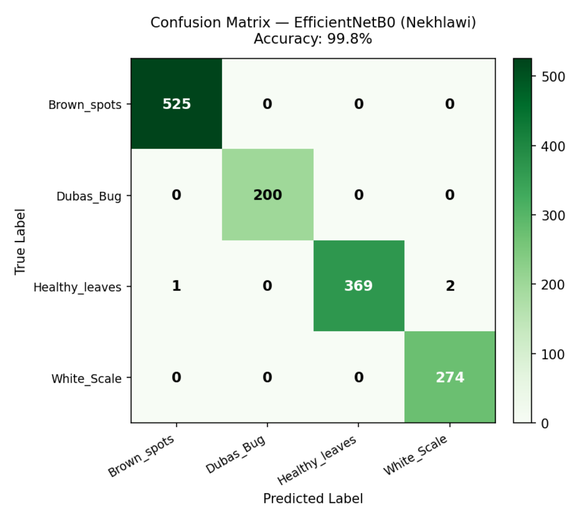
Nekhlawi is a one-stop AI-based mobile platform that fills the gap between the traditional farming and modern technology by: using AI-Driven Diagnosis, Expert Connectivity.

7 Results

- Developed Nekhlawi, an AI-based application that detects palm diseases through image analysis and deep learning
- Creating a hub for accessing knowledge and having a connection with experts
- Provide a solution that supports digital transformation in agriculture and enhances sustainability

- Trained the EfficientNetB0 AI model on computer vision and classification to diagnose date palm tree diseases which reached the optimal accuracy of 99.8%
- Implemented GRAD CAM to provide the end-user model transparency and explainable AI





8 Conclusion

Nekhlawi proposes a solution to reduce the time for detecting palm diseases, improving the future of agricultural technology and minimizing losses of efforts and money through image processing. With the integration of the knowledge hub and experts' connection, empowering users' decision-making.

9 Tools